

Code Quality & Security Audit Report

Comprehensive Static Code & Dependency Analysis

PROJECT	DART SDK	TARGET PLATFORMS	STATE / NETWORK
flutter_base	^3.4.4	android, ios	setState / dio

1. Executive Summary & Category Breakdown

CATEGORY	HIGH SEVERITY	MEDIUM SEVERITY	LOW SEVERITY	TOTAL ISSUES
Security & Secrets	33	0	1	34
Performance & UI Lag	23	0	0	23
Memory & Lifecycle Leaks	37	0	0	37
Maintainability & Class Size	1	44	0	45
Total Detected Issues	94	44	1	139

2. Critical Security Findings

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
SEC-SEC-001	CRITICAL	ios/Runner/AppDelegate.swift	12	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** <code>dart const String stripeApiKey = "sk_live_51H...";</code> <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	ios/Runner/GoogleService-Info_old.plist	6	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** <code>dart const String stripeApiKey = "sk_live_51H...";</code>

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				<i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	android/app/google-services_old.json	18	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	android/app/src/main/AndroidManifest.xml	81	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>

Rule ID	Severity	File Location	Line	Finding Details & Risk
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	169	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	191	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	191	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	198	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	199	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	203	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	207	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ```

RULE ID	SEVERITY	FILE LOCATION	LINE	FINDING DETAILS & RISK
				<i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	221	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	221	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>

Rule ID	Severity	File Location	Line	Finding Details & Risk
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	222	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-SEC-001	CRITICAL	lib/utils/common/constants.dart	222	Hardcoded API Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	164	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	165	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application

Rule ID	Severity	File Location	Line	Finding Details & Risk
				package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	169	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	191	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	198	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they

Rule ID	Severity	File Location	Line	Finding Details & Risk
				can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	199	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/common/constants.dart	520	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const String stripeApiKey = "sk_live_51H..."; ``` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-ATH-001	HIGH	lib/utils/helpers/shared_pref_helper.dart	12	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ```dart const

Rule ID	Severity	File Location	Line	Finding Details & Risk
				String stripeApiKey = "sk_live_51H..."; ```` Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
SEC-ATH-001	HIGH	lib/utils/helpers/shared_pref_helper.dart	22	Potential Hardcoded Password or Auth Key: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ````dart const String stripeApiKey = "sk_live_51H..."; ```` Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
SEC-WV-001	HIGH	lib/reusable_component/cms_page.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ````dart const String stripeApiKey = "sk_live_51H..."; ```` Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
SEC-WV-002	HIGH	lib/reusable_component/reusables.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ````dart const String stripeApiKey = "sk_live_51H..."; ````

Rule ID	Severity	File Location	Line	Finding Details & Risk
				<i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-WV-002	HIGH	lib/screens/account_tab/dpharma_plus/dpharma_plus_plans.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H..."; `` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-WV-002	HIGH	lib/screens/cart/cart_list/my_cart_bloc.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H..."; `` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-WV-001	HIGH	lib/screens/cart/payment_webview/payment_webview.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H..."; `` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend</i>

Rule ID	Severity	File Location	Line	Finding Details & Risk
				databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.
SEC-WV-002	HIGH	lib/screens/cart/payment_webview/payment_webview.dart	1	Unsafe WebView Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H...";`` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-AND-003	HIGH	android/app/src/main/AndroidManifest.xml	1	Android Insecure Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H...";`` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-IOS-001	HIGH	ios/Runner/Info.plist	1	iOS Insecure Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** ``dart const String stripeApiKey = "sk_live_51H...";`` <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>

Rule ID	Severity	File Location	Line	Finding Details & Risk
SEC-IOS-002	HIGH	ios/Runner/Info.plist	1	iOS Insecure Configuration: Hardcoded credentials are static values that get compiled directly into the application package. If someone decompiles or reverse-engineers the application, they can recover these secrets. **Bad Example:** <code>dart const String stripeApiKey = "sk_live_51H...";</code> <i>Risk: Exposing sensitive API keys or credentials can allow unauthorized access to backend databases, server integrations, cloud infrastructure, or user data, leading to severe security breaches and financial liabilities.</i>
SEC-DEP-INFO	LOW	pubspec.yaml	1	Vulnerability Database Check Status: Using built-in offline Dart/Flutter security rules database. Remote live vulnerability database check is not configured. <i>Risk: None. This is an informational message stating that live CVE lookup from OSV/GitHub Advisory is not active.</i>

Security Remediation Action Items

- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. ****Good Example:**** `dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key');`
- Periodically check dependencies using GitHub security alerts or running "dart pub token" or external tools.

3. High-Priority Performance Flaws: Controller in build()

Instantiating controllers inside build() forces memory re-allocations on every frame render, causing UI stutter, dropped text inputs, and lost scroll states.

Component / Area	File Path	Line	Issue Description
Pharmacylistandmap	lib/screens/NearByPharmacy/pharmacylistandmap.dart	142	Controller Instantiation Inside Widget build() Method
Chat Details	lib/screens/chat/details/chat_details.dart	252	Controller Instantiation Inside Widget build() Method

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
Tabbar	lib/screens/tabbar/tab/tabbar.dart	70	Controller Instantiation Inside Widget build() Method
Searched Products Pharm List	lib/screens/search/searched_products_pharm_list/searched_products_pharm_list.dart	43	Controller Instantiation Inside Widget build() Method
Home	lib/screens/home_tab/home/home.dart	105	Controller Instantiation Inside Widget build() Method
Featured Brand List	lib/screens/home_tab/featured_brands/featured_brand_list.dart	35	Controller Instantiation Inside Widget build() Method
Product List	lib/screens/home_tab/product_list/product_list.dart	62	Controller Instantiation Inside Widget build() Method
Address List	lib/screens/home_tab/address_list/address_list.dart	94	Controller Instantiation Inside Widget build() Method
Select Map Location	lib/screens/home_tab/address_list/select_map_location.dart	79	Controller Instantiation Inside Widget build() Method
Product Other Sellers	lib/screens/home_tab/product_details/product_other_sellers/product_other_sellers.dart	41	Controller Instantiation Inside Widget build() Method
Product Review List	lib/screens/home_tab/product_details/review_list/product_review_list.dart	49	Controller Instantiation Inside Widget

COMPONENT / AREA	FILE PATH	LINE	ISSUE DESCRIPTION
			build() Method
My Messages List	lib/screens/account_tab/my_messages/my_messages_list.dart	58	Controller Instantiation Inside Widget build() Method
Refer Friend	lib/screens/account_tab/refer_friend/refer_friend.dart	53	Controller Instantiation Inside Widget build() Method
My Prescriptions	lib/screens/account_tab/my_prescriptions/list/my_prescriptions.dart	40	Controller Instantiation Inside Widget build() Method
Manage Address	lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart	37	Controller Instantiation Inside Widget build() Method
Order Ticket List	lib/screens/account_tab/order_related_queries/order_ticket_list.dart	60	Controller Instantiation Inside Widget build() Method
My Wishlist	lib/screens/account_tab/my_wishlist/my_wishlist.dart	39	Controller Instantiation Inside Widget build() Method
Notification List	lib/screens/account_tab/notifications/notification_list.dart	43	Controller Instantiation Inside Widget build() Method
My Cart	lib/screens/cart/cart_list/my_cart.dart	124	Controller Instantiation Inside Widget build() Method

Component / Area	File Path	Line	Issue Description
Pres Order List	lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart	92	Controller Instantiation Inside Widget build() Method
My Orders	lib/screens/my_orders_tab/orders/list/my_orders.dart	55	Controller Instantiation Inside Widget build() Method
Select Order	lib/screens/my_orders_tab/create_new_ticket/select_order.dart	39	Controller Instantiation Inside Widget build() Method
Category List	lib/screens/category_tab/category_list/category_list.dart	44	Controller Instantiation Inside Widget build() Method

Standard Fix Architecture:

```
class _SafeInputWidgetState extends State<SafeInputWidget> {
  late final TextEditingController _controller;

  @override
  void initState() {
    super.initState();
    _controller = TextEditingController(); // Allocated once
  }

  @override
  void dispose() {
    _controller.dispose(); // Safely freed from memory
    super.dispose();
  }

  @override
  Widget build(BuildContext context) => TextField(controller: _controller);
}
```

4. Memory & Resource Leaks

Issue Type	File Path	Line	Hazard
MEM-001 (HIGH)	lib/utils/helpers/push_notification_helper.dart	441	Unclosed StreamSubscription

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-002 (HIGH)	lib/utils/helpers/pref_navigator_observer.dart	65	Unclosed Timer
MEM-003 (HIGH)	lib/screens/NearByPharmacy/pharmacylistandmap_bloc.dart	16	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/order_via_prescription/upload_prescription/upload_prescription.dart	32	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/order_via_prescription/upload_prescription/upload_prescription_bloc.dart	19	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/auth/verify_otp/verify_otp_bloc.dart	22	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/auth/complete_profile/complete_profile.dart	30	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/auth/complete_profile/complete_profile_bloc.dart	20	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/auth/login/login_bloc.dart	22	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/auth/login/login.dart	29	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/search/search_list/search_list_bloc.dart	22	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/home_tab/featured_brands/featured_brand_list.dart	23	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/home_tab/featured_brands/featured_brand_list_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/home_tab/search_addresses/search_addresses_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/home_tab/product_list/product_list_bloc.dart	48	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/my_messages/my_messages_list_bloc.dart	19	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/refer_friend/refer_friend_bloc.dart	15	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/edit_profile/edit_profile_bloc.dart	20	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/edit_profile/edit_profile.dart	35	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/manage_address/save_address/save_address_bloc.dart	39	AnimationController Not Disposed

ISSUE TYPE	FILE PATH	LINE	HAZARD
MEM-003 (HIGH)	lib/screens/account_tab/manage_address/save_address/save_address_bloc.dart	28	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart	26	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/account_tab/order_related_queries/order_ticket_list_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/cart_list/my_cart.dart	51	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/coupons/coupon_list_bloc.dart	15	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/cart/coupons/coupon_list.dart	27	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/order_details/order_details_order_placed.dart	1112	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart	35	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_product/rate_product_bloc.dart	21	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_product/rate_product.dart	30	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/orders/list/my_orders_bloc.dart	14	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_seller/rate_seller.dart	30	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/rate_seller/rate_seller_bloc.dart	19	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/create_new_ticket_bloc.dart	17	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/select_order_bloc.dart	14	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/my_orders_tab/create_new_ticket/create_new_ticket.dart	39	AnimationController Not Disposed
MEM-003 (HIGH)	lib/screens/category_tab/category_list/category_list_bloc.dart	16	AnimationController Not Disposed

5. Class Bloat & Technical Debt (Files > 500 LOC)

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
lib/utils/common/constants.dart	1108	Utility Helpers	Split large files into smaller focused units
lib/utils/helpers/push_notification_helper.dart	682	Utility Helpers	Split large files into smaller focused units
lib/reusable_component/toast/flushbar.dart	793	General	Split large files into smaller focused units
lib/reusable_component/reusables.dart	1187	General	Split large files into smaller focused units
lib/screens/NearByPharmacy/pharmacylistandmap.dart	639	UI / View	Split large files into smaller focused units
lib/screens/order_via_prescription/upload_prescription/upload_prescription.dart	606	UI / View	Split large files into smaller focused units
lib/screens/chat/details/chat_details.dart	741	UI / View	Split large files into smaller focused units
lib/screens/search/search_list/search_list.dart	722	UI / View	Split large files into smaller focused units
lib/screens/home_tab/home/home.dart	1337	UI / View	Split large files into smaller focused units
lib/screens/home_tab/product_list_filter/product_list_filter.dart	665	UI / View	Split large files into smaller focused units
lib/screens/home_tab/product_list_filter/product_list_filter_bloc.dart	670	UI / View	Split large files into smaller focused units
lib/screens/home_tab/search_addresses/search_addresses_bloc.dart	-	UI / View	Save references to all subscriptions created with <code>`.listen()`</code> and invoke <code>`.cancel()`</code> inside <code>`.dispose()`</code> for Widgets, or <code>`.onClose()`</code> for GetxControllers. **Good Example:** <code>dart class _MyState extends State { late StreamSubscription _subscription; @override void initState() { super.initState(); _subscription = myStream.listen((data) { setState(() {}); }); } @override void dispose() { _subscription.cancel(); super.dispose(); } }</code>
lib/screens/home_tab/product_list/product_list.dart	732	UI / View	Split large files into smaller focused units

FILE LOCATION	LINES	CATEGORY / FOCUS AREA	ARCHITECTURAL ACTION
lib/screens/home_tab/address_list/address_list.dart	830	UI / View	Split large files into smaller focused units
lib/screens/home_tab/product_details/details/product_details.dart	1285	UI / View	Split large files into smaller focused units
lib/screens/account_tab/edit_profile/edit_profile.dart	591	UI / View	Split large files into smaller focused units
lib/screens/account_tab/manage_address/save_address/save_address.dart	852	UI / View	Split large files into smaller focused units
lib/screens/account_tab/profile_list/profile_list.dart	523	UI / View	Split large files into smaller focused units
lib/screens/cart/cart_list/my_cart.dart	1908	UI / View	Split large files into smaller focused units
lib/screens/cart/cart_list/my_cart_bloc.dart	632	UI / View	Split large files into smaller focused units
lib/screens/my_orders_tab/order_details/order_details_order_placed.dart	1717	UI / View	Split large files into smaller focused units
lib/screens/my_orders_tab/pres_order_list/pres_order_list.dart	911	UI / View	Split large files into smaller focused units
lib/screens/my_orders_tab/orders/filter/my_orders_filter.dart	515	UI / View	Split large files into smaller focused units

6. Unresolved Production TODOs

FILE PATH	LINE	MISSING FEATURE / INCOMPLETE CODE
lib/reusable_component/pdf_view.dart	25	Declared widget/class/function has no references
lib/reusable_component/pinch_zoom_page.dart	26	Declared widget/class/function has no references
lib/screens/NearByPharmacy/pharmacylistandmap.dart	51	Declared widget/class/function has no references
lib/screens/splash/intro_page.dart	33	Declared widget/class/function has no references
lib/screens/auth/complete_profile/complete_profile.dart	38	Declared widget/class/function has no references
lib/screens/search/searched_products_pharm_list/searched_products_pharm_list.dart	29	Declared widget/class/function has no references
lib/screens/search/Pharmacy_products_list_add_cart/Pharmacy_products_list_add_cart.dart	33	Declared widget/class/function has no references

FILE PATH	LINE	MISSING FEATURE / INCOMPLETE CODE
lib/screens/home_tab/home/home_bloc.dart	40	Declared widget/class/function has no references
lib/screens/home_tab/product_list_filter/product_list_filter_bloc.dart	441	Declared widget/class/function has no references
lib/screens/home_tab/featured_brands/featured_brand_list.dart	28	Declared widget/class/function has no references
lib/screens/home_tab/search_addresses/search_addresses.dart	32	Declared widget/class/function has no references
lib/screens/home_tab/address_list/select_map_location.dart	46	Declared widget/class/function has no references
lib/screens/home_tab/product_details/details/product_details.dart	37	Declared widget/class/function has no references
lib/screens/home_tab/product_details/product_other_sellers/product_other_sellers.dart	27	Declared widget/class/function has no references
lib/screens/home_tab/product_details/review_list/product_review_list.dart	32	Declared widget/class/function has no references
lib/screens/account_tab/refer_friend/refer_friend.dart	30	Declared widget/class/function has no references
lib/screens/account_tab/manage_address/manage_addresses/manage_address.dart	30	Declared widget/class/function has no references
lib/screens/account_tab/profile_list/profile_list.dart	43	Declared widget/class/function has no references
lib/screens/my_orders_tab/rate_product/rate_product_bloc.dart	26	Declared widget/class/function has no references
lib/screens/my_orders_tab/rate_product/rate_product.dart	35	Declared widget/class/function has no references
lib/screens/my_orders_tab/rate_seller/rate_seller.dart	35	Declared widget/class/function has no references
lib/screens/my_orders_tab/rate_seller/rate_seller_bloc.dart	24	Declared widget/class/function has no references

7. Recommended Action Plan

Phase 1 — Hotfix (Days 1–2):

- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:**

```
dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key');
```
- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:**

```
dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key');
```
- Remove hardcoded credentials from code. Use environment variables during the build process, or retrieve keys dynamically from a secure vault or backend service at runtime. **Good Example:**

```
dart import 'package:flutter_secure_storage/flutter_secure_storage.dart'; final storage = FlutterSecureStorage(); final stripeKey = await storage.read(key: 'stripe_api_key');
```

Phase 2 — Performance & Memory (Days 3–5):

- Convert the widget to a `StatefulWidget` and initialize the controller inside `initState()`. Ensure you also call `.dispose()` on it inside the `dispose()` override. **Good Example:**

```
dart class MyInputWidget extends StatefulWidget { const MyInputWidget({super.key}); @override State createState() => _MyInputWidgetState(); } class _MyInputWidgetState extends State { late final TextEditingController _textController; @override void initState() { super.initState(); _textController = TextEditingController(); @override void dispose() { _textController.dispose(); super.dispose(); } @override Widget build(BuildContext context) { return TextField(controller: _textController); } }
```
- Convert the widget to a `StatefulWidget` and initialize the controller inside `initState()`. Ensure you also call `.dispose()` on it inside the `dispose()` override. **Good Example:**

```
dart class MyInputWidget extends StatefulWidget { const MyInputWidget({super.key}); @override State createState() => _MyInputWidgetState(); } class _MyInputWidgetState extends State { late final TextEditingController _textController; @override void initState() { super.initState(); _textController = TextEditingController(); @override void dispose() { _textController.dispose(); super.dispose(); } @override Widget build(BuildContext context) { return TextField(controller: _textController); } }
```
- Save references to all subscriptions created with `.listen()` and invoke `.cancel()` inside `dispose()` for Widgets, or `onClose()` for GetxControllers. **Good Example:**

```
dart class _MyState extends State { late StreamSubscription _subscription; @override void initState() { super.initState(); _subscription = myStream.listen((data) { setState(() {}); }); @override void dispose() { _subscription.cancel(); super.dispose(); } }
```

Phase 3 — Refactoring & Debt (Week 2):

- Split large files into smaller focused units
- Split large files into smaller focused units
- Address pending TODO comments in `lib/reusable_component/pdf_view.dart` at line 25.

8. Analysis Criteria Observed

The project is audited against the following rules:

Code Quality	Security
Proper use of const and final: If widget not updated once declare and variable value will never change once declare then use const and final with decoration (Threshold: <code>const_possible == true</code> <code>assignment_count == 1</code>) Large File LOC: Trigger when a file has more than 500 non-empty, non-comment code lines (Threshold: <code>file_code_loc > 500</code>) Large Class LOC: Trigger when a class exceeds configured LOC threshold (Threshold: <code>class_code_loc > 300</code>) Large Function LOC: Trigger when a function/method exceeds configured LOC threshold (Threshold: <code>function_code_loc > 80</code>) Unused Widgets/Classes/Functions: Declared widget/class/function has no references (Threshold: <code>reference_count == 0</code>) Unused Imports: Imported symbol is never referenced (Threshold: <code>import_reference_count == 0</code>) Unreachable Branch: Branch can never execute based on static control-flow analysis (Threshold: <code>unreachable == true</code>) High Cyclomatic Complexity: Function exceeds complexity threshold (Threshold: <code>complexity > 10</code>) Duplicate Code: Similar code block exceeds configured similarity threshold (Threshold: <code>similarity >= 0.85</code>) Large Build Method: Build method exceeds configured size (Threshold: <code>build_loc > 80</code>) Deep Widget Nesting: Widget tree nesting exceeds threshold (Threshold: <code>depth > 8</code>) Broad State Rebuild: State update potentially rebuilds unnecessarily large widget subtree (Threshold: <code>Configurable</code>) Missing Const Opportunity: Constructor/widget can safely be const but is not (Threshold: <code>const_possible == true</code>)	Hardcoded API Key: API key pattern detected in source/config (Threshold: <code>Pattern + entropy score</code>) Payment Gateway Credential: Payment provider private/secret credential detected (Threshold: <code>Provider pattern</code>) Firebase Service Account: Firebase service-account JSON/private key detected (Threshold: <code>Credential pattern</code>) Private Key/Certificate: PEM/private key/signing material detected (Threshold: <code>private_key_detected == true</code>) Insecure Token Storage: Access/refresh tokens stored in SharedPreferences/plain files (Threshold: <code>Sensitive token + insecure storage</code>) HTTP/Cleartext Traffic: Non-TLS HTTP endpoint or cleartext platform configuration detected (Threshold: <code>http:// / cleartext enabled</code>) Certificate Validation Bypass: Client bypasses certificate validation (Threshold: <code>Trust-all callback/API</code>) Unsafe WebView Configuration: JavaScript/file access/untrusted navigation enabled without controls (Threshold: <code>Risk pattern</code>) Unsafe Dynamic Data Handling: Unsafe eval-like/dynamic deserialization or execution pattern detected (Threshold: <code>Risk API pattern</code>) Sensitive Data in Logs: Token/password/payment/PII passed to logging APIs (Threshold: <code>Sensitive variable/pattern</code>) Android Insecure Configuration: Debuggable release config, exported components or permissive network settings (Threshold: <code>Risk configuration</code>) iOS Insecure Configuration: ATS exceptions, insecure entitlements or overly permissive settings (Threshold: <code>Risk configuration</code>) Overly Permissive Firebase Rules: Firestore/Realtime Database allows broad unauthenticated read/write (Threshold: <code>Broad read/write access</code>)

- **Improper final Usage:** Local variable never reassigned and can be final (Threshold: `assignment_count == 1`)
- **Unsafe Late Usage:** late variable may be accessed before initialization or is unnecessary (Threshold: `Configurable`)
- **Naming Convention Violation:** Classes, variables, methods/files violate Dart naming conventions (Threshold: `Dart convention`)
- **Empty Catch Block:** Catch block has no meaningful handling/logging/rethrow (Threshold: `catch_body_effective_statements == 0`)

Memory Leak

- **Unclosed StreamSubscription:** StreamSubscription created without matching cancellation/disposal (Threshold: `cancel_missing == true`)
- **Unclosed Timer:** Timer created without cancellation (Threshold: `cancel_missing == true`)
- **AnimationController Not Disposed:** AnimationController created in State without dispose (Threshold: `dispose_missing == true`)
- **TextEditingController Not Disposed:** TextEditingController retained without disposal (Threshold: `dispose_missing == true`)
- **Scroll/Page Controller Not Disposed:** ScrollController/PageController created without disposal (Threshold: `dispose_missing == true`)
- **Listener Not Removed:** addListener/addObserver without matching remove (Threshold: `remove_missing == true`)
- **Unnecessary StatefulWidget:** StatefulWidget has no mutable state or lifecycle requirement (Threshold: `stateful_need == false`)
- **Missing Const StatelessWidget:** Immutable widget does not use const constructor (Threshold: `const_possible == true`)
- **Long-Lived Closure Captures State/Context:** Long-lived callback captures State/BuildContext (Threshold: `Risk pattern`)
- **Unbounded In-Memory Collection:** List/map/cache grows without size/eviction control (Threshold: `Configurable size/eviction rule`)

Dependencies

- **Vulnerable Package Version:** Direct/transitive dependency has known security advisory (Threshold: `Known CVE/advisory`)
- **Unused Dependency:** Declared package has no project references (Threshold: `reference_count == 0`)
- **Unmaintained Package:** Package has stale releases or weak maintenance signals (Threshold: `Configurable inactivity period`)

Reporting

- **File/Line/Code Evidence:** Every finding must include exact source location and evidence snippet (Threshold: `Required fields`)
- **Severity/Confidence Scoring:** Every finding receives severity and confidence (Threshold: `Required fields`)
- **Suppression/Ignore Support:** Approved findings can be suppressed with a reason (Threshold: `Suppression requires reason`)

Runtime Analysis

- **Heap Growth Over Time:** Heap increases across repeated flows and does not return toward baseline (Threshold: `Configurable growth %`)
- **Widget/Object Retention:** Objects remain retained after route/widget disposal (Threshold: `Retained after disposal`)
- **Frame Rendering/Jank:** Frame build/raster time exceeds threshold or dropped frames detected (Threshold: `frame_time > 16.67ms`)
- **Excessive Widget Rebuilds:** Widget rebuild count significantly exceeds expected baseline (Threshold: `Configurable multiplier`)
- **Slow Startup:** Cold-start time exceeds configured threshold (Threshold: `startup_ms > 3000`)
- **Slow API Request:** API request exceeds latency threshold (Threshold: `latency_ms > 2000`)
- **Excessive Network Traffic:** Unexpected request frequency or payload volume (Threshold: `Configurable request/payload threshold`)
- **Unhandled Exception:** Unhandled Dart/Flutter exception during test flow (Threshold: `unhandled_exception == true`)
- **Resource Retained After Screen Exit:** Subscription/timer remains active after navigation away (Threshold: `Active resource after dispose`)
- **Sensitive Runtime Storage:** Token/payment/PII written to logs or insecure storage (Threshold: `Sensitive data detected`)

Architecture

- **Layer Violation:** UI directly accesses infrastructure/data layer against selected architecture (Threshold: `Forbidden dependency`)
- **Mixed State Management:** Conflicting state-management patterns used across modules (Threshold: `Configured allowed patterns`)
- **Business Logic Inside Widget:** Networking/business rules embedded in build/UI classes (Threshold: `Configurable complexity/API count`)

Local Threshold Parameters

- **Max Lines of Code:** 500
- **Max Function Lines:** 50
- **Allow TODO:** false
- **Enforce class prefixes:** true